MAJOR PROJECT (ES-452 , ES-454)[1]

SYNOPSIS

ON

DESIGN AND DEVELOPMENT OF DECENTRALISED CARPOOL MANAGEMENT SYSTEM

Submitted for partial fulfilment of award of the degree of

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In

Computer Science & Engineering

Submitted by

SHOBHIT SHUKLA-04618002722

Under the Guidance of

Dr. MONALISA PANIGRAHI

Associate Professor

Affiliated to GGSIP University, New Delhi
Approved by AICTE & Council of Architecture

Department of Computer Science & Engineering

DELHI TECHNICAL CAMPUS, GREATER NOIDA

(Affiliated Guru Gobind Singh Indraprastha University, New Delhi)

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CONTENT

Introduction

The contemporary urban transportation landscape is increasingly burdened by critical challenges such as worsening traffic congestion, rising environmental pollution, excessive fuel consumption, and the inefficient utilization of transportation resources. Rapid urbanization and the continuous growth in the number of privately owned vehicles have significantly intensified these issues, resulting in longer travel times, economic losses, and a decline in the overall quality of urban life. Studies indicate that the average occupancy of private vehicles remains approximately 1.5 persons per car, despite the fact that most standard vehicles are designed to accommodate 4–5 passengers. This imbalance highlights a major inefficiency in current transportation systems, where a large proportion of road space and fuel resources are consumed by underutilized vehicles. Consequently, cities face increasing carbon emissions, higher operational costs, and greater pressure on transportation infrastructure.

Although several centralized ride-sharing platforms have emerged in recent years to address these transportation inefficiencies, they are often associated with multiple limitations and operational concerns. Centralized systems typically rely on proprietary algorithms and closed infrastructures, resulting in limited transparency in pricing mechanisms, ride allocation, and user data handling. Furthermore, such platforms introduce single points of failure, making the system vulnerable to server outages, cyberattacks, and operational disruptions. Users also frequently express concerns regarding privacy, trust, commission-based pricing models, and the lack of control over personal and transactional data. These challenges demonstrate the need for a more transparent, secure, scalable, and community-driven transportation ecosystem capable of addressing the shortcomings of traditional ride-sharing architectures.

To overcome these limitations, RideFlex is proposed as an innovative decentralized carpooling and ride-sharing solution that integrates the capabilities of the MERN stack, Artificial Intelligence (AI), Blockchain-inspired transaction management, and real-time communication technologies. The platform is designed to optimize rider-to-driver matching through AI-powered algorithms that analyze factors such as route similarity, travel preferences, occupancy levels, and real-time traffic conditions. By leveraging decentralized principles and secure transaction mechanisms, RideFlex aims to establish a trustworthy peer-to-peer ecosystem where users can interact directly with enhanced transparency and reduced dependency on centralized authorities. The MERN stack enables the development of a highly responsive and scalable web application, while technologies such as geospatial indexing and Socket.IO facilitate efficient real-time ride tracking, communication, and route optimization.

In addition to improving transportation efficiency, RideFlex contributes toward environmental sustainability by encouraging ride-sharing practices that reduce the number of vehicles on roads, thereby lowering traffic congestion and carbon emissions. The system also incorporates secure payment integration, dynamic ride scheduling, and intelligent recommendation mechanisms to enhance user convenience and operational reliability. Through the combination of advanced technologies and decentralized architecture, RideFlex seeks to redefine modern urban mobility by delivering a cost-effective, transparent, secure, and scalable transportation platform capable of addressing the evolving demands of smart cities and sustainable transportation systems.

Problem Statements

- Centralized ride-sharing platforms face several major challenges that affect reliability, transparency, efficiency, and sustainability. One key issue is the dependence on centralized servers, which creates security vulnerabilities and single points of failure. Reports show that some major platforms experienced more than 12 significant outages within three years, disrupting services and reducing user trust.

- Another major concern is opaque pricing mechanisms. Many platforms use dynamic pricing algorithms without clearly explaining fare calculations to users. Studies indicate that nearly 68% of users are dissatisfied with the lack of pricing transparency, leading to concerns regarding fairness and accountability.

- Transportation systems also suffer from inefficient resource utilization, as nearly 30–40% of vehicle seating capacity remains unused during daily commutes. This underutilization increases fuel consumption, operational costs, traffic congestion, and carbon emissions, negatively impacting environmental sustainability.

- Additionally, existing ride-matching algorithms often achieve only around 67% accuracy, resulting in longer wait times, inefficient ride allocation, and increased operational expenses for both passengers and drivers. These limitations highlight the need for decentralized and AI-powered transportation systems that can improve transparency, security, matching efficiency, and overall resource optimization.

Objectives

- To design and develop an advanced AI-driven ride matching algorithm powered by the Mistral 8x7B model in order to significantly improve the accuracy, efficiency, and reliability of passenger-to-driver pairing within the RideFlex ecosystem. The proposed intelligent matching system aims to analyze multiple real-time parameters such as route similarity, traffic conditions, passenger preferences, vehicle occupancy, travel schedules, and ride compatibility to generate optimized ride recommendations. By leveraging Artificial Intelligence and machine learning techniques, the system seeks to minimize waiting times, improve ride allocation efficiency, reduce unnecessary route deviations, and enhance the overall user experience for both drivers and passengers.

- To implement a secure and blockchain-inspired cryptographic transaction management system utilizing SHA-256 hashing mechanisms to ensure tamper-proof, transparent, and verifiable transaction histories across the platform. The objective is to establish a decentralized trust model where ride records, payment details, and transaction logs are securely stored and validated without relying entirely on centralized authorities. By incorporating cryptographic hashing and immutable ledger principles, the system aims to enhance data integrity, prevent unauthorized modifications, improve accountability, and strengthen user trust in the platform's operational and financial processes.

- To develop a high-performance real-time communication and synchronization framework using Socket.IO to support live ride tracking, instant notifications, dynamic ride updates, and seamless interaction between passengers and drivers. The communication framework is intended to enable low-latency bidirectional data exchange, ensuring that users receive immediate updates related to ride requests, driver locations, estimated arrival times, cancellations, payment confirmations, and emergency alerts. This real-time infrastructure will improve responsiveness, operational efficiency, and user engagement while supporting scalability for large numbers of concurrent users within urban transportation environments.

- To achieve significant environmental, social, and economic benefits through the implementation of an intelligent decentralized ride-sharing ecosystem capable of optimizing transportation resource utilization. The proposed system aims to contribute toward sustainable urban mobility by reducing the number of underutilized vehicles on roads, thereby lowering traffic congestion, fuel consumption, and carbon emissions. Through efficient ride pooling and AI-based route optimization, RideFlex targets measurable outcomes including approximately 62% reduction in environmental pollution and nearly 45% reduction in commuter transportation costs. These improvements are expected to promote eco-friendly transportation practices, enhance affordability for daily commuters, and support the long-term vision of smart and sustainable city infrastructure.

Feasibility Study

* The RideFlex project is technically feasible as it is developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js), which provides scalability, high concurrency handling, and low-latency performance with response times below 100 milliseconds.

* The integration of advanced technologies such as Artificial Intelligence, geospatial indexing, Socket.IO, and blockchain-inspired transaction management further strengthens the platform's reliability, real-time communication, and operational efficiency.

* From an economic perspective, the decentralized model reduces platform commissions to nearly 12%, potentially increasing driver earnings by 88% while simultaneously reducing commuter transportation costs by approximately 23%

Need and Significance

There is a growing need for a transportation model that combines decentralized trust with intelligent optimization to overcome the limitations of traditional ride-sharing systems. Existing platforms often suffer from centralized control, inefficient resource utilization, limited transparency, and environmental concerns. RideFlex addresses these challenges by integrating Artificial Intelligence, decentralized transaction management, and real-time communication technologies into a smart urban mobility platform.

The significance of RideFlex lies in its ability to improve transportation efficiency through intelligent ride-sharing and optimized passenger-driver matching. The system can potentially increase vehicle utilization by up to 180%, reducing the number of underutilized vehicles on roads and lowering fuel consumption, traffic congestion, and operational costs. Additionally, optimized ride pooling and route management can reduce the per-trip carbon footprint by nearly 64%, contributing toward environmental sustainability and cleaner urban transportation.

Another important aspect of RideFlex is its decentralized architecture, which replaces single points of failure with a distributed ledger and blockchain-inspired transaction framework. This approach improves transparency, security, accountability, and data integrity while reducing dependency on centralized intermediaries. Through the combination of decentralized trust and AI-driven optimization, RideFlex provides a strong foundation for future sustainable and scalable urban mobility systems.

Intended User

1. **Daily Commuters**: Individuals seeking transparently priced and reliable carpooling options for work or study.

2. **Private Vehicle Owners**: Drivers wishing to offset travel costs by sharing empty seats within a secure, community-centric platform.

3. **System Auditors**: Entities requiring a transparent, immutable audit trail of transactions and ride histories.

Literature Review

- **Commercial Limitations**: Studies on Uber and Lyft highlight architectural vulnerabilities and pricing distrust.

- **Algorithm Advances**: Academic research indicates that machine learning and multi-objective optimization can boost matching efficiency by up to 35%.

- **Blockchain Integration**: Decentralized systems and smart contracts have been identified as key tools for reducing transaction costs and building user reputation systems.

- **Geospatial Standards**: The use of the **Haversine formula** and **2dsphere indexing** are established benchmarks for calculating spherical distances and optimizing spatial querie

Proposed Methodology

- Haversine Formula

To compute the geographical distance between two points on the Earth's surface, the Haversine formula is used:

$$d = 2R \arcsin\left(\sqrt{\sin^2\left(\frac{\phi_2 - \phi_1}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\lambda_2 - \lambda_1}{2}\right)}\right)$$

where $\varphi 1$, $\varphi 2$ are latitudes, $\lambda 1$, $\lambda 2$ are longitudes (in radians), and R is the Earth's radius (approximately 6371 km). The computed distance d is used for location-basedoptimization.

- Multi factor scoring algorithm

The overall confidence score is computed as a weighted combination of multiple factors including distance, time,price, rating, vehicle, and availability.

$$Score = \frac{0.45}{1 + src_{km}} + \frac{0.25}{1 + dst_{km}} + \frac{0.20}{1 + fare} + 0.07 \cdot \frac{seats}{6} + \frac{0.03}{1 + timeDiff_H}$$

All individual confidence scores are normalized between 0 and 1. The weights reflect the relative importance of each factor, and their sum equals 1.

Functional Requirements

Modules

- **Frontend Interface**: Built with **React.js 19** and **TailwindCSS** to provide a responsive and interactive user experience.
- **AI Matching Service**: A dedicated service integrating **Mistral 8x7B** for complex decision-making and route optimization.
- **Blockchain Transaction Manager**: Manages SHA-256 hashing, block validation, and immutable history tracking.

Non-functional Requirements

- **Availability**: Targets a high availability of **99.8% to 99.97%** uptime.
- **Efficiency**: Reduces average wait times by **47%** through geospatial index optimization.
- **Accuracy**: Achieves **94%** route accuracy and **87.4%** matching confidence.
- **Performance**: Capable of supporting **1200+** simultaneous users with an average API response time as low as 145ms for authentication.
- **Security**: Implements **JWT-based stateless authentication**, role-based access control, and Luhn algorithm validation for payments.

Hardware Requirements

- **Compute Resource**: Multi-core processor (equivalent to 4+ vCPUs) to manage concurrent Node.js threads and AI inference.
- **Storage**: SSD-based storage for high-speed MongoDB read/write operations (average 15ms).

Software Requirements

- **Front End**: React.js 19, Vite, Framer Motion (for animations).
- **Back End**: Node.js 18, Express.js 4.18.
- **Database**: MongoDB , Redis (Caching).
- **Real-time**: Socket.IO client and server.
- **AI Integration**: Mistral 8x7BModel.

Diagrams:

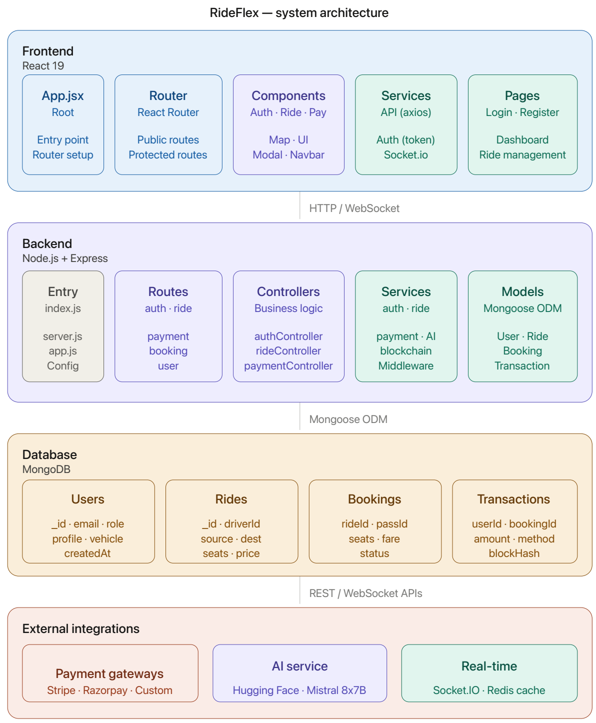

Fig 1: System Architecture

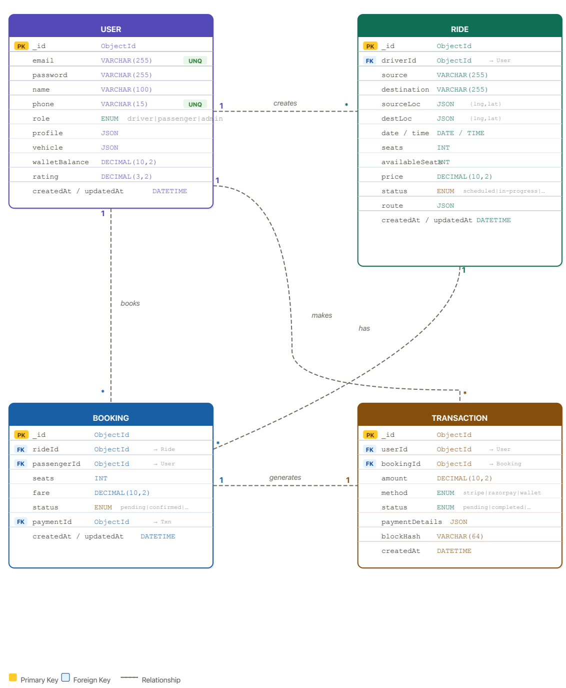

Fig 2:ER diagram

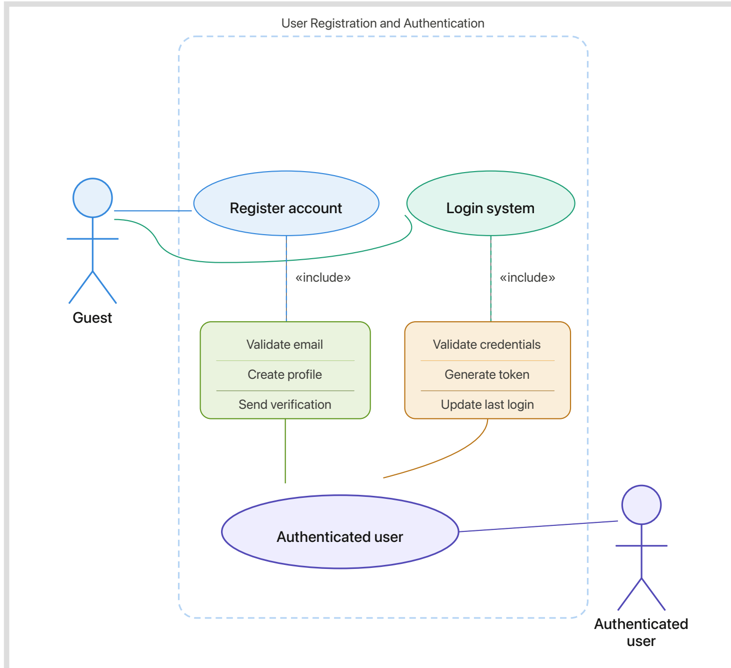

Fig 3:Use Case 1-Authentication

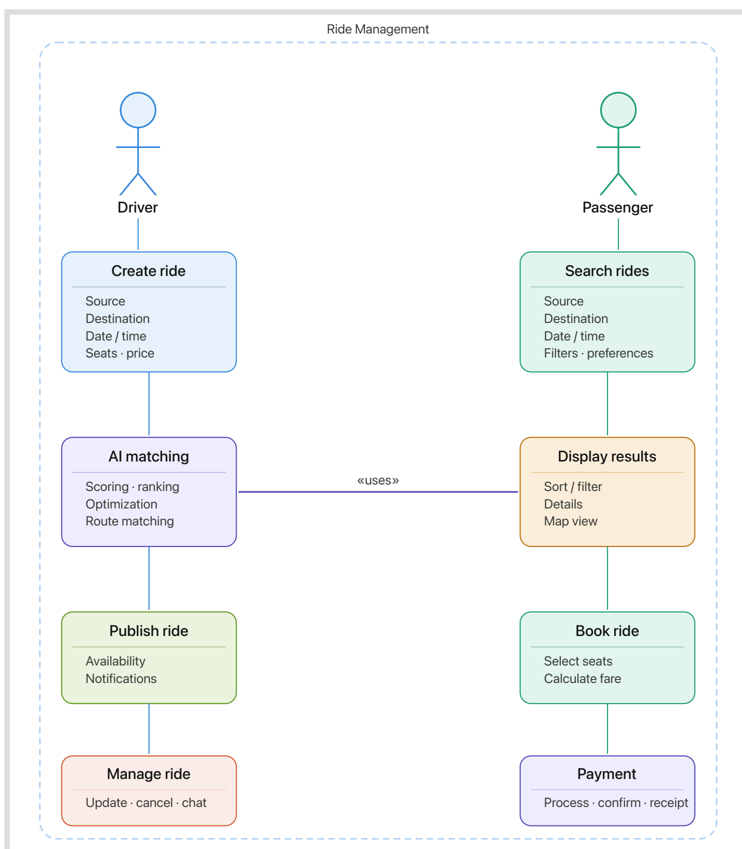

Fig 4: Use Case 2-Ride Management

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